Justin Valentine

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EDUCATION

University of Alberta — Edmonton, AB
MSc in Computing Science (thesis), GPA 4.0/4.0, supervised by Martin Jagersand — *Sep 2024 – Dec 2026*
BSc, double major in Mathematics and Computing Science — *Sep 2019 – Dec 2023*

EXPERIENCE

Barrett Technology — Greater Boston, MA
Haptic wrist design, firmware and controls — *May – Sep 2026*

- Engineered the Alberta wrist end to end, a 3-DOF cable-driven haptic wrist that makes a 4-DOF WAM a 7-DOF teleoperation leader arm, from cardboard prototypes to production drawings in under three months.
- Ran FEA, worked through cable stiffness, resonance, reflected inertia and motor selection, and worked with machine shops on tolerances. Both wrists reach 33–40 Hz closed-loop bandwidth per joint.
- Ported the WAM to CANopen, from bare-metal C on the safety controller to real-time C++ control.
- Automated a motor dynamometer for efficiency, thermal and peak-torque tests, and wrote Python tools for Puck motor control, gravity calibration, dynamics identification and friction compensation.
- Built a 500 Hz haptic lightsaber game, with a Meta Quest 3 viewer, for Barrett's IROS 2026 booth.

University of Alberta — Edmonton, AB
Graduate Research Assistant, Computer Vision and Robotics Group — *Sep 2024 – Present*

- Designed and built a series of haptic wrists for teleoperating Barrett WAM arms, four of them cable driven, and a PS5-trigger handle PCB. Machined over a dozen of their parts; led three undergraduates.
- Supervised two MecE capstone teams building cable-driven haptic trigger handgrips; one won best capstone.
- WHAM wrist: a 2-DOF cable-driven wrist with a passive joint, from mobility analysis, so the operator's arm sets the WAM's pose, one-way clutch bearings, and cable tensioners built into the pinions. Paper in preparation.
- Co-developed the lab's C++ teleoperation stack (libhaptic_wrist, wam_teleop): wrist kinematics, gravity calibration, an orientation controller that clears the direct-drive wrist's singularity, and joystick control of follower J7.
- Wrote Robotiq and moteus gripper drivers in libgripper, fed gripper force back to the trigger, and built a 500 Hz test suite.

Teaching Assistant, Computer Vision; Introduction to Robotics and Mechatronics — *Jan – May 2025*
Undergraduate Research Assistant, Computer Vision and Robotics Group — *Dec 2022 – Sep 2024*

- Designed the base, sensors and ROS setup of a haptic mobile manipulator (a WAM on a Clearpath Husky with a 360° camera streamed to the operator's VR headset), one of eight finalists in DND's IDEaS challenge.
- Built a BLDC motor test stand on a Teensy 4.1, and worked on wrist mechanisms with Johns Hopkins.

Millennium Three Technologies *Mechanical design and manufacturing* — *Sep 2024 – Present*

- Designed and machined aluminum parts that made a box-picking robot stiffer and lighter.

PROJECTS

- **HAL quadruped** (2023–24). A Solo 8-style quadruped whose motorized spine changes its pitch inertia in flight to attempt a backflip. Milled the chassis and 24 pulleys; training it in MuJoCo MJX with Brax PPO.
- **Variable-pitch hopper** (2026). A one-legged hopper whose optimized screw-cam groove sets its gear ratio. Its 1 kHz controller, built in MuJoCo, runs in C++ on a custom Teensy 4.1 board, checked tick by tick against Python.
- **Sketch generation with flow matching** (2024). Generates sketches point by point with flow matching and classifier-free guidance; a WAM then draws them on a whiteboard that presses back with a constant force.

SELECTED PUBLICATIONS

- A. Noohian, D. Miller, **J. Valentine**, A. Lynch, M. Jagersand. "Sensorless Four-Channel Control Architecture Using Inverse Dynamics Modeling for Human-Scale Bilateral Teleoperation." Submitted to IROS 2026.
- A. Wang, C. Jiang, M. Przystupa, **J. Valentine**, M. Jagersand. "Point and Go: Intuitive Reference Frame Reallocation in Mode Switching for Assistive Robotics." ICRA 2025. **Best Student Paper**; nominated for Best HRI Paper.
- S. Akturk, **J. Valentine**, J. Ahmad, M. Jagersand. "Immersive Human-in-the-Loop Control: Real-Time 3D Surface Meshing and Physics Simulation." IROS 2024.
- **J. Valentine**, D. Miller, F. Haghverd, J. Piwowarczyk, M. Fiala, M. Jagersand. "Increasing Situational Awareness in Robotic Manipulation: A High-Force Haptic System." Poster, RISE 2025.

SKILLS

Software: C++, Python, C, Swift, MATLAB, CUDA; ROS, libbarrett, MuJoCo/MJX, Brax, PyTorch, Isaac Sim, OpenCV
Embedded and control: real-time control, CANopen, Teensy firmware, BLDC drives, haptic rendering, KiCad
Mechanical: SolidWorks, Fusion 360, FEA, cable drives, CAD/CAM, CNC milling and turning, waterjet